STAFF HEALTH RISK EXPOSURE & BUILDING SAFETY REPORT

ISO 14971 & IPC — Is the Old Dental Clinic Building safe for staff to work in, now and over the years?

Old Dental Clinic, VNH | 28 Oct 2026 | Routine review

Assessor	S. Kalki	**Clinic / IPC Lead**	—
Building age / year	—	**Next review due**	—
Areas assessed	Operatories / Waiting / Sterilization / Lab / Plant Rooms		
Staff exposed	12 staff · 40 h/week	**Years occupied / planned**	8 / 5
Overall compliance	**87.8%**	**Exposure Index (0-100)**	8.3

STAFF SAFETY VERDICT — CURRENT

UNSAFE — restricted use until controls in place

LONG-TERM (MULTI-YEAR OCCUPANCY)

NOT ACCEPTABLE for long-term occupancy — 2 chronic exposure hazard(s) open (A2, C7)

BASIS FOR STAFF SAFETY VERDICT

- 5 High-risk gap(s): A2, A11, B8, C7, E3
- Long-term: NOT ACCEPTABLE for long-term occupancy — 2 chronic exposure hazard(s) open (A2, C7)

EXECUTIVE SUMMARY

Staff health risk exposure assessment of Old Dental Clinic, VNH (Routine review), 28 Oct 2026, by S. Kalki. Purpose: determine whether the building is safe and healthy for staff to work in, now and over continued years of occupancy.

- Items assessed: 41 of 41
- Overall compliance: 87.8% (Yellow; target >= 90%)
- Staff Exposure Index: 8.3 / 100 (lower = healthier)
- Gaps: 5 (High 5 · Medium 0 · Low 0)
- Chronic (cumulative) exposure hazards open: 2 (A2, C7)
- Change since 15 Sept 2024: compliance +12.2 pts; Exposure Index 12.6 -> 8.3; verdict Unsafe (do not occupy) -> Unsafe (restricted)

STAFF EXPOSURE PROFILE

- Staff working in building: 12
- Average time in building: 40 h/week
- Years occupied to date: 8; planned further occupancy: 5 years
- Building: age not recorded; areas: Operatories / Waiting / Sterilization / Lab / Plant Rooms

LONG-TERM (MULTI-YEAR) EXPOSURE

Open hazards where harm accumulates with each year staff work in the building:

- A2 No visible water damage, dampness, or active leaks in walls, ceilings, floors... — NOT MET [score 6]: Damp -> mould spores; allergic & respiratory disease with prolonged exposure
- C7 Mercury / amalgam waste is managed according to environmental regulations — NOT MET [score 6]: Mercury vapour -> neurological & renal toxicity (cumulative)

Cumulative exposure projection (indicative)

- Now: 8 yrs in building · 96 staff-years exposed · 96 hazard-score-years per staff — Very high cumulative load

— relocation / major remediation warranted
- +1 year: 9 yrs in building · 108 staff-years exposed · 108 hazard-score-years per staff — Very high cumulative load — relocation / major remediation warranted
- +5 years: 13 yrs in building · 156 staff-years exposed · 156 hazard-score-years per staff — Very high cumulative load — relocation / major remediation warranted
- +10 years: 18 yrs in building · 216 staff-years exposed · 216 hazard-score-years per staff — Very high cumulative load — relocation / major remediation warranted

STAFF HEALTH SURVEILLANCE

- Period: 12 months; 12 incidents / symptom reports; rate 10 per 10 staff per year
- Sharps / needlestick injuries: 1
- Respiratory complaints (cough, asthma, wheeze): 2
- Building-related symptoms (headache, eye / throat irritation, fatigue): 9
- HEALTH SIGNAL: 92% of staff report building-related, respiratory or skin symptoms — investigate building as a cause
- Health surveillance programme: not recorded; documented HBV immunity: not recorded

DOMAIN RESULTS

- A. Building Infrastructure: 10/12 compliant (83.3%) — Yellow
- B. Biological / IPC: 9/10 compliant (90%) — Green
- C. Chemical Hazards: 6/7 compliant (85.7%) — Yellow
- D. Physical & Equipment: 6/6 compliant (100%) — Green
- E. Ergonomic & Organisational: 5/6 compliant (83.3%) — Yellow
- OVERALL: 36/41 (87.8%) — Yellow

KEY FINDINGS — HAZARDS TO STAFF

High risk — immediate control

- A2 NOT MET: No visible water damage, dampness, or active leaks in walls, ceilings, floors or plant rooms [S2 x P3 = 6] (chronic)
- A11 NOT MET: Fire detection, alarm and extinguisher systems are inspected and functional [S2 x P3 = 6] (critical)
- B8 NOT MET: Reusable instruments are cleaned, packaged and sterilised according to manufacturer IFU [S2 x P3 = 6]
- C7 NOT MET: Mercury / amalgam waste is managed according to environmental regulations [S2 x P3 = 6] (chronic)
- E3 NOT MET: Staff know how to report hazards, near-misses and incidents [S2 x P3 = 6]

RECOMMENDATIONS (HIERARCHY OF CONTROLS)

- PRIORITY: restrict or suspend staff work in affected areas until critical / high-risk controls are verified.
- Remediate chronic hazards first — every additional year of occupancy adds to staff cumulative exposure. If remediation is not feasible, plan relocation.
- Refer symptomatic staff to occupational health; correlate symptoms with building areas.

Engineering

- A2: Repair leaks and dry affected materials within 48 h; remove water-damaged porous materials
- A11: Service fire systems; clear exits; install exit signage; schedule fire drills
- C7: Sealed amalgam waste containers; licensed disposal; amalgam separator check

Administrative

- B8: Weekly spore testing with log; IFU-compliant cycles; validate steriliser
- E3: Launch non-punitive incident reporting; brief staff; review reports monthly

CONCLUSION

The building is assessed as: UNSAFE — restricted use until controls in place. Long-term: NOT ACCEPTABLE for long-term occupancy — 2 chronic exposure hazard(s) open (A2, C7). Implement the corrective action plan and

re-assess to confirm the building is safe for continued staff occupancy.

Next review due: —

The completed checklist, domain scoring, staff health data, multi-year trend and corrective action plan are attached as Annex A.

SIGN-OFF

Assessor:
Date: —

Clinic / IPC Lead:
Date: —

ANNEX A — STAFF HEALTH RISK EXPOSURE CHECKLIST & DATA

Dental Staff Health Risk Assessment — Old Dental Clinic Building · Biological · Chemical · Physical · Building Infrastructure · Ergonomic

1. Assessment Information

Facility / Clinic Name	Old Dental Clinic, VNH	**Date**	28 Oct 2026
Assessor Name / Role	S. Kalki	**Building Age / Year Built**	
Areas Assessed	Operatories / Waiting / Sterilization / Lab / Plant Rooms	**Next Review Due**	—

2. Risk Rating Method (ISO 14971-aligned)

Score	Severity of Harm	Probability / Likelihood	Risk Level & Action
1 - Low	Minor injury / temporary discomfort	Rare / unlikely	Monitor; routine controls
2 - Medium	Reversible injury / illness requiring treatment	Possible / occasional	Corrective action within defined timeframe
3 - High	Serious / irreversible injury, chronic disease, or death	Likely / frequent	Immediate control measures required

Risk score = Severity x Probability (1-9): Low 1-2 · Medium 3-4 · High 6-9. Domain % = Yes / (Yes + No). Green >= 90% · Yellow 70-89% · Red < 70%.

3. Checklist Domains

A. Building Infrastructure & Environmental Health (Old Building Specific)

83.3% · Yellow

#	Checklist Item	Primary Risk	Yes	No	N/A	S	P	Score	Notes / Evidence
A1	Building condition survey / structural assessment has been completed within the last 5 years (or as required by local regulation)	Physical [Critical]	X						
A2	No visible water damage, dampness, or active leaks in walls, ceilings, floors or plant rooms	Biological / Physical [Chronic]		X		2	3	**6 High**	
A3	No evidence of mould / fungal growth; any historical mould remediation is documented and verified	Biological [Chronic] [Critical]	X						
A4	Asbestos-containing materials (if present in older buildings) have been surveyed, labelled and managed under an asbestos management plan	Chemical / Physical [Chronic] [Critical]	X						
A5	Lead-based paint (common in pre-1978 buildings) has been assessed; no deteriorating paint in staff or patient areas	Chemical [Chronic] [Critical]	X						
A6	HVAC / ventilation system is adequate for clinical and laboratory areas; air-change rates meet current standards; filters cleaned/replaced on schedule	Biological / Chemical [Chronic] [Critical]	X						
A7	Local exhaust ventilation (LEV) or dust extraction is functional at high-dust workstations (lab, polishing)	Chemical / Physical [Chronic]	X						
A8	Dental unit waterlines (DUWL) are treated and monitored; water quality meets <=500 CFU/mL (or local standard); biofilm control program in place	Biological [Chronic]	X						
A9	Plumbing is free of significant corrosion or stagnation risks; dead-legs minimised; legionella risk assessment performed if required	Biological [Chronic]	X						
A10	Electrical systems (wiring, panels, outlets) have been inspected; no overloaded circuits, exposed wiring or damaged outlets	Physical [Critical]	X						
A11	Fire detection, alarm and extinguisher systems are inspected and functional; emergency exits clear and signed	Physical [Critical]		X		2	3	**6 High**	
A12	Pest control program is active; no evidence of rodents or insects that could contaminate clinical areas	Biological	X						

B. Biological Hazards & Infection Prevention and Control (IPC)

90% · Green

#	Checklist Item	Primary Risk	Yes	No	N/A	S	P	Score	Notes / Evidence
B1	Written IPC / Exposure Control Plan is current, site-specific, and reviewed at least annually	Biological	X						
B2	All staff with occupational exposure have received bloodborne pathogen training (initial + annual) and HBV vaccination is offered/documented	Biological	X						
B3	Appropriate PPE (gloves, masks, eye protection, protective clothing) is available, used correctly, and removed before leaving clinical/lab areas	Biological	X						
B4	Hand-hygiene facilities (sinks with soap/paper towels or alcohol-based hand rub) are available at point of care and stocked	Biological	X						
B5	Impressions, prostheses and appliances are cleaned and disinfected before laboratory processing and before return to the patient	Biological	X						
B6	Clean and contaminated zones are clearly separated; workflow prevents cross-contamination	Biological	X						
B7	Clinical contact surfaces are cleaned and disinfected between patients with EPA-registered (or equivalent) intermediate-level disinfectant; contact time observed	Biological	X						
B8	Reusable instruments are cleaned, packaged and sterilised according to manufacturer IFU; biological monitoring performed at least weekly	Biological		X		2	3	6 High	
B9	Sharps containers are puncture-resistant, labelled, not overfilled, and located near point of use	Biological / Physical	X						
B10	Post-exposure management protocol (needlestick / splash) is written, accessible, and staff know how to activate it	Biological	X						

C. Chemical Hazards

85.7% · Yellow

#	Checklist Item	Primary Risk	Yes	No	N/A	S	P	Score	Notes / Evidence
C1	Safety Data Sheets (SDS) are available and accessible for all chemicals used (disinfectants, acrylics, monomers, cleaners, amalgam, etc.)	Chemical	X						
C2	All chemical containers are labelled with contents and hazard warnings; secondary containers also labelled	Chemical	X						
C3	Flammable chemicals stored in approved cabinets; incompatible chemicals segregated	Chemical	X						
C4	Adequate ventilation / LEV controls chemical vapours and dust (acrylic, gypsum, metals, disinfectants)	Chemical [Chronic]	X						
C5	Spill kits and emergency eyewash / shower are available, inspected and functional	Chemical	X						
C6	Hazard Communication (HazCom) training completed by all relevant staff	Chemical	X						
C7	Mercury / amalgam waste is managed according to environmental regulations; no open amalgam scrap	Chemical [Chronic]		X		2	3	6 High	

D. Physical Hazards & Equipment Safety

100% · Green

#	Checklist Item	Primary Risk	Yes	No	N/A	S	P	Score	Notes / Evidence
D1	Dental chairs, lights, handpieces and other equipment are in safe working order; maintenance logs current	Physical	X						
D2	Grinding / polishing equipment (if lab present) has functioning guards, shields and dust collection	Physical	X						
D3	Eye protection with side shields (or face shields)	Physical	X						

#	Checklist Item	Primary Risk	Yes	No	N/A	S	P	Score	Notes / Evidence
	is worn during procedures generating debris or splash								
D4	Floors, walkways and stairs are free of trip/slip hazards; adequate lighting in all work areas	Physical	X						
D5	Noise levels from equipment are assessed; hearing protection available where needed	Physical [Chronic]	X						
D6	Compressed gas cylinders (if used) are secured and stored correctly	Physical	X						

E. Ergonomic & Organisational Controls

83.3% · Yellow

#	Checklist Item	Primary Risk	Yes	No	N/A	S	P	Score	Notes / Evidence
E1	Workstations allow neutral posture; adjustable seating and adequate task lighting are provided	Ergonomic [Chronic]	X						
E2	Job rotation or micro-breaks are used for repetitive or sustained tasks	Ergonomic [Chronic]	X						
E3	Staff know how to report hazards, near-misses and incidents; reporting system is non-punitive	Organisational		X		2	3	**6 High**	
E4	Written IPC and occupational health & safety policies are available, current and communicated to staff	Organisational	X						
E5	Staffing levels and workload are reviewed for fatigue-related risk	Organisational [Chronic]	X						
E6	Emergency action plan (fire, medical emergency, exposure incident) is written, practised and known to staff	Organisational	X						

4. Domain Summary Scoring (ICAR-style)

Domain	Yes	No	N/A	% Compliant	Priority
A. Building Infrastructure	10	2	0	83.3%	**Yellow**
B. Biological / IPC	9	1	0	90%	**Green**
C. Chemical Hazards	6	1	0	85.7%	**Yellow**
D. Physical & Equipment	6	0	0	100%	**Green**
E. Ergonomic & Organisational	5	1	0	83.3%	**Yellow**
OVERALL	**36**	**5**	**0**	**87.8%**	**Yellow**

Priority key: Green >= 90% | Yellow 70-89% | Red < 70%. High-severity gaps (Score 3) require immediate action regardless of domain percentage.

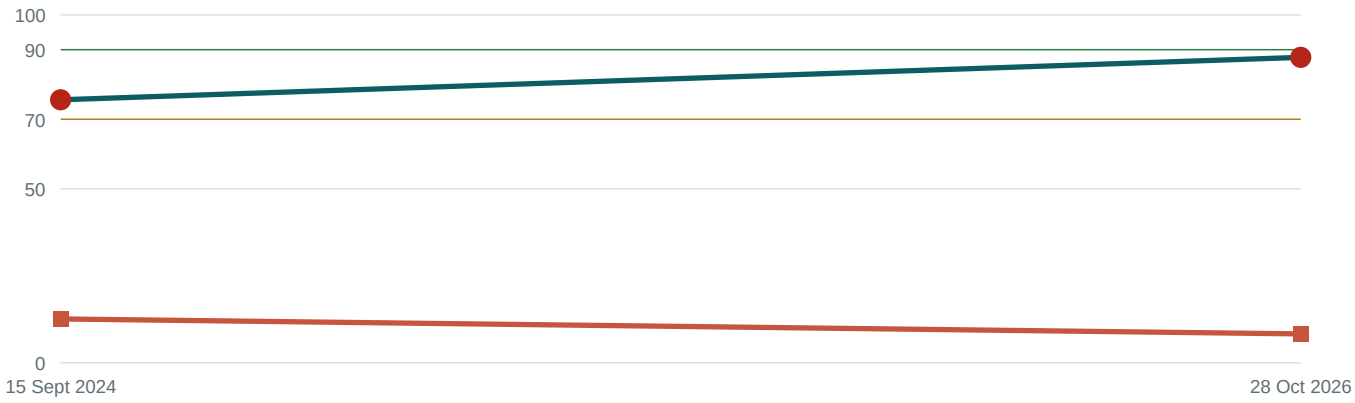
5. Staff Exposure Profile & Health Surveillance

Measure	Value (period 12 months)
Staff working in building	12
Average hours / week in building	40
Years occupied to date	8
Planned further years of occupancy	5
Staff groups exposed	—
Health surveillance programme	—
Documented HBV immunity (%)	—
Sharps / needlestick injuries	1
Blood / body-fluid splash exposures	0

Measure	Value (period 12 months)
Work-acquired infections (suspected)	0
Chemical exposure incidents / burns	0
Respiratory complaints (cough, asthma, wheeze)	2
Building-related symptoms (headache, eye / throat irritation, fatigue)	9
Skin / dermatitis complaints	0
Musculoskeletal complaints (back, neck, wrist)	0
Hearing complaints / tinnitus	0
Slips, trips, falls, other injuries	0
Sick-leave days attributed to work / building	0
Incident / symptom rate per 10 staff per year	10
Staff with building-related symptoms (%)	92%
Health signal (>= 20%)	YES

6. Multi-Year Trend

Date	Type	Compliance	Exposure Index	High risks	Chronic open	Incidents /10 staff/yr	Current verdict	Long-term
15 Sept 2024	Baseline	75.6%	12.6	7	3	5.8	Unsafe (do not occupy)	Not acceptable
28 Oct 2026	Routine review	87.8%	8.3	5	2	10	Unsafe (restricted)	Not acceptable



Line teal = Compliance % (dot colour = verdict) | Line orange = Exposure Index (lower = healthier) | 90% target, 70% red line

7. Cumulative Chronic Exposure Projection (indicative)

Horizon	Years in building	Staff-years exposed	Hazard-score-years / staff	Interpretation
Now	8	96	96	Very high cumulative load — relocation / major remediation warranted
+1 year	9	108	108	Very high cumulative load — relocation / major remediation warranted
+5 years	13	156	156	Very high cumulative load — relocation / major remediation warranted
+10 years	18	216	216	Very high cumulative load — relocation / major remediation warranted

#	Open chronic hazard	Score	Long-term health effect
A2	No visible water damage, dampness, or active leaks in walls, ceilings, floors...	6	Damp -> mould spores; allergic & respiratory disease with prolonged exposure
C7	Mercury / amalgam waste is managed according to	6	Mercury vapour -> neurological & renal toxicity (cumulative)

#	Open chronic hazard	Score	Long-term health effect
	environmental regulations		

8. Corrective Action Plan

#	Gap / Non-compliance	Control Measure	Hierarchy	Responsible	Target Date	Status
1						
2						
3						
4						
5						

Hierarchy of controls: 1. Eliminate -> 2. Substitute -> 3. Engineering controls -> 4. Administrative controls -> 5. PPE (last line of defence).

9. Key References (APA 7th)

Australian Dental Association. (n.d.). Guidelines for infection prevention and control (5th ed., amended).

Centers for Disease Control and Prevention. (2003). Guidelines for infection control in dental health-care settings—2003. MMWR Recommendations and Reports, 52(RR-17), 1–61.

Centers for Disease Control and Prevention. (n.d.). Infection Control Assessment and Response (ICAR) tool for general infection prevention and control (IPC) across settings. https://www.cdc.gov/healthcare-associated-infections/php/toolkit/icar.html

International Organization for Standardization. (2019). Medical devices — Application of risk management to medical devices (ISO 14971:2019).

Occupational Safety and Health Administration. (n.d.). Bloodborne pathogens standard (29 CFR 1910.1030) and Hazard communication standard (29 CFR 1910.1200).

Public Health Ontario. (2019). IPAC checklist for dental practices – Core elements (1st revision).

World Health Organization. (2018). Infection prevention and control assessment framework at the facility level (WHO/HIS/SDS/2018.9).